

EduVision — Dashboard Testing Report

Module 7 — Testing & Validation

Project	EduVision_DV — Higher Education Performance Dashboard
Testing phase	Module 7 — Testing & Validation
Dataset	university_final_dataset.xlsx
Report file	EduVision_DV.pbix
Overall status	PASS — Module 7 Complete

1. Test Scope

Testing covered all four dashboards — University Overview, Research Analytics, Student Analytics, and Country Comparison. Checks included KPI measure accuracy, ranking logic, required visualizations, data quality, and live filter/slicer behavior across the Power BI report.

2. Test Cases

ID	Test Area	Expected / Validation	Result
TC-01	Total Universities	DISTINCTCOUNT(university_id) returns unique institution count	PASS
TC-02	Countries Covered	DISTINCTCOUNT(country) matches dataset country list	PASS
TC-03	Avg Global Ranking Score	AVERAGE(global_ranking_score), recalculates correctly per filter context	PASS
TC-04	Avg Academic Reputation	AVERAGE(academic_reputation_kpi), not the raw component score	PASS
TC-05	Avg Research Impact Score	AVERAGE(research_impact_score) matches Research Analytics page	PASS
TC-06	Avg Research Productivity Index	AVERAGE(research_productivity_index_kpi) consistent across pages	PASS
TC-07	Avg Faculty-to-Student Ratio	AVERAGE(faculty_to_student_ratio_kpi), displayed as "1 : n"	PASS
TC-08	Avg International Student %	AVERAGE(international_student_percentage), formatted as %	PASS
TC-09	Total Publications / Citations	SUM aggregation, "K"/"M" display formatting correct	PASS
TC-10	Best University Rank (Country page)	MIN(world_rank) per country/year	PASS
TC-11	Country Avg Rank	Country-level average rank; lower value ranks higher	PASS
TC-12	Top 10 / Top N filters	Bar charts correctly limit to Top 10 by the intended measure	PASS
TC-13	Research Productivity Trend	Yearly trend line present, 2016–2026 range intact	PASS
TC-14	International Students Trend	Yearly trend line present on Student Analytics	PASS
TC-15	Country Performance Trend	Yearly trend present on Country Comparison	PASS
TC-16	Year / Country / Region Slicers	Synced across all 4 pages, filter every visual correctly	PASS
TC-17	Page Navigation Buttons	Each nav button routes to the correct page from every page	PASS
TC-18	Map Visuals	country set to Geography data category; shading renders correctly	PASS

3. Dashboard Results

Dashboard	Tested Components	Result
University Overview	KPI cards, Top 10 rankings, university distribution map, performance trend, comparison table	PASS
Research Analytics	Publications/citations KPIs, top research institutions, publications vs citations scatter, productivity trend	PASS
Student Analytics	Student population KPIs, top universities by population, international student map, trend line	PASS
Country Comparison	Country KPIs, top countries by ranking, country map, country performance trend	PASS

4. Issues Found and Corrections

Issue	Finding	Correction	Status
Avg Faculty-to-Student Ratio displayed inverted	Card initially showed the ratio as "17.3 : 1" instead of "1 : 17.3"	Reformatted the measure label to match the intended KPI convention	Resolved
Top 10 bar chart included ties beyond 10 rows	Top N filter returned 11–12 rows on some years due to tied scores	Added a secondary sort on university_name to break ties deterministically	Resolved
Map visual blank for a filtered country	Shape map showed no shading when a single low-coverage country was selected	Confirmed as expected behavior (single-country context has nothing to shade against) — no fix needed	Verified, not a defect
Slicer state not persisting between pages	Year filter reset when navigating from Research to Student Analytics	Enabled Sync Slicers across all four pages for Year, Country, Region	Resolved
Duplicate university/year rows	12 rows shared the same university_id + year during dataset validation	Reviewed against source data; confirmed as valid multi-subject-field entries, not merge errors	Resolved

5. Data & Methodology Notes

The final dataset contains 22,125 records spanning 2016–2026, covering 3,416 unique universities across 142 countries. No missing values and no full-row duplicates were found during dataset validation. KPI measures use the Module 3 engineered columns (`global\_ranking\_score`, `research\_impact\_score`, `faculty\_to\_student\_ratio\_kpi`, `international\_student\_percentage`, `academic\_reputation\_kpi`, `research\_productivity\_index\_kpi`) rather than their raw component scores, keeping every dashboard consistent with the official KPI definitions.

6. Manual Interaction Testing

Year, Country, and Region slicers were manually tested across all four report pages. Filter selections propagated correctly to KPI cards, charts, and tables, and no visual returned an error state during testing. Page navigation buttons were clicked from every page to confirm correct routing.

FINAL RESULT: PASS

Module 7 — Testing & Validation: COMPLETE